Forensic Lens: Deepfake Detection Through Micro-Level Facial Blood-Flow Signals

Abstract—**This research introduces a novel, computationally efficient deepfake detection method that employs a Forensic Lens to analyze biological authenticity in video content. The core concept leverages remote photoplethysmography (rPPG) signals to identify the natural, rhythmic color variations on human facial skin caused by blood pressure changes from the cardiac cycle. These periodic "glow" effects are typically absent or irregular in synthetically generated deepfake videos. Using the Celeb-DF V2 dataset, the method isolates facial regions, extracts rPPG signals from the green color channel dynamics, and utilizes a label-based classification approach. The proposed model achieves an overall accuracy of 90% with high precision (0.88), recall (0.89), and F1-score (0.88). This performance demonstrates that biological consistency analysis is a powerful, explainable, and trustworthy cue for deepfake detection, offering a significant alternative to visually- or texture-based methods. Crucially, the model's minimal computational complexity and low latency make it highly suitable for real-time applications and deployment on resource-constrained edge devices, providing a practical and physiologically grounded solution to the escalating deepfake threat.**

I. INTRODUCTION

Artificial images are produced by deepfake technology. It is generated with the help of deep learning (DL) that focuses on learning the features of a specific image and traces it to another image. In this way creation of forged images is performed successfully. This phenomenon has gained popularity on applications like Snapchat, Bigo, TikTok, etc. [53]. These technologies provide threats in areas like politics, entertainment, and cybersecurity, but they have also brought up serious ethical and security issues in addition to providing previously unheard-of creative possibilities. Because deepfakes are frequently indistinguishable from real media, they are frequently used maliciously for impersonation, disseminating false information, and other purposes. The use of extremely lifelike synthetic media made possible by recent developments in deepfake technology has increased the number of fraud cases [6]. It is used by criminals to exchange fake ones with real data. The initial acceptance of deepfakes in entertainment and creative fields has transformed into a serious challenge against digital media privacy and security as well as trust within the digital domain [55]. In 2024, for instance [54]. A deepfake video of Elon Musk (last viewed on February 17, 2025) was used to advertise a cryptocurrency fraud, tricking an 82-year-old retiree into investing 690,000, which ultimately led to a total loss of money. In 2024, cyber fraud also made rapid use of AI technologies [6]. The research topic examines deepfake digital forensics through an initiative to create better detection methods for this field. Deepfakes have emerged as a vital research subject because they are increasingly misused by attackers who create false media to harm in various ways including spreading misinformation while robbing people financially and stealing their identity [42]. identity theft and the dissemination of false information on social media. The academic community has become very interested in these deepfakes. Deepfakes have sparked worries about innocent people's safety and security during their brief life [53]. Digital integrity, along with individual protection and technology ethics, depend on immediate solutions to resolve this issue. This research advances the knowledge in cyber-security alongside artificial intelligence by closing present forensic gaps that allow society to address deepfake threats. Deepfake technology has become widespread in our society, thus making this research field highly significant and relevant [61]. Deepfakes have eroded the trust of the media. Deepfakes contribute to misinformation, privacy breaches, and increased cybercrime. Audio and deepfakes enable scams and manipulate public sentiment, harming society. [9], [55]. Protecting information requires efficient deepfake detection at a time when digital media functions as the main source of information across the global network [18].Traditional deepfake detection techniques are becoming increasingly ineffective due to the improved sophistication of these algorithms. Media authenticity relies and effective forensic tools to detect and safeguard against deepfakes. The examination of this area leads to enhanced digital safety along with ethical and moral behavior and promotes disinformation defense, which supports more dependable and safe online systems [45]. Deepfakes require urgent investigation into digital forensics because they present multiple essential problems. The main challenge circulating today revolves around the advanced evolution of deepfake algorithms [18]. GAN-based techniques develop modern media with such high fidelity that traditional detection capabilities become useless. As the real media becomes indistinguishable from generated content [32]. The current detection methods face challenges when applying them to scale up according to modern needs. Modern digital forensics detection tools demand powerful computing systems that reduce their operational effectiveness in emergency security situations [1]. The methods have limited ability to recognize different deepfake manipulation types because they show poor data generalization performance across diverse datasets. The field faces ethical obstacles which affect its operation [31]. Privacy guidelines, consent-related requirements, data safeguards, and security processes are the main issues while doing forensic examination of sensitive material. Maintaining strong detection capabilities while regarding ethical concerns presents a fundamental problem to solve [18].Deepfake misapplications create worsened problems in society including false information propagation and online harassment together with unauthorized personal information theft. Such malicious deepfake applications including the generation of fake political statements and non-consensual pornography create significant consequences for both people and communities [9]. This research develops new forensic methods for addressing the existing digital security issues in order to create a safer digital environment. The development of deepfake technologies stems from artificial intelligence algorithms, specifically through Generative Adversarial Networks (GANs). The creation of artificial data through GANs became possible after Ian Goodfellow introduced these networks in 2014 [19]. The new content generation method, which began in 2014, led to deepfakes becoming more prevalent throughout society. The initial research centered on artistic and entertainment applications of GANs until deepfake misuse received widespread attention [31]. At first, scientists were interested in using visual cues like excessive blinking and odd lighting as indicators for further research. Deepfake algorithms underwent advancements that minimized the usefulness of these(Error-Inducing) cues as markers of synthetic content [56]. Research now studies sophisticated detection systems which analyze the mismatches between facial expressions and voice patterns and biological human signals. Heart rate estimation methods in DeepRhythm, along with subtle changes in facial color detection through DeepFakesON-Phys work to expose the manipulation of digital contents [10]. Researchers have developed AI-based classifiers and machine learning models that have enhanced the identification of deepfake alterations. The research makes progress by implementing physiological signal analysis together with leading forensic instruments to overcome previous scaling and generalizing obstacles [32]. In order to provide digital forensics capabilities that combat deepfakes, the research integrates historical insights with contemporary technology advancements. Introduction of deepfake technology together with forensic research affects three groups: both individual users and organizational entities as well as governmental bodies. Deepfakes mainly target people who are public figures because they frequently experience damaging consequences through privacy violations as well as reputation destruction along with identity theft incidents [9]. Regular platform users face vulnerability because deepfakes contribute to false information distribution which leads to exploitation. Deepfake technology creates substantial impacts that affect primarily organizations functioning within cybersecurity and media and legal fields [59].Media organizations need to develop strategies to stop disinformation from spreading and cybersecurity businesses need to acquire improved technologies to defend their clientele from deepfake-facilitated financial scams. Legal institutions encounter obstacles when trying to establish proof of evidence credibility during deepfake examinations [55]. Governments together with their policymakers hold essential roles in this matter. Professionals in law enforcement require dependable forensic methods to fight against threats to national security, manage election interference and stabilize social environments degraded by deepfake technology. This research produces results which protect the trust and security and maintain accountability throughout the digital age for all stakeholders. There are essential understanding and detection limitations regarding deepfakes despite recent advancements. The detection techniques now in use have difficulty achieving both large-scale applicability across several datasets and data diversity stability [41]. Specific deepfake training algorithms demonstrate limited effectiveness in real-life scenarios because they lose detection capabilities when detecting new manipulations which they have never encountered [40]. The difficulty arises from the fact that forensic techniques are not integrated with physiological data, such as blood pressure and heart rate readings. Research development needs additional refinement to optimize the detection methods despite their promising results especially when handling variations in video quality and noise factors originating from the environment [47]. Studies concerning the ethical dimensions of deepfake forensics have not yet received enough investigation. The study focuses on filling the research gaps surrounding forensic tool privacy along with misuse by creating detection methods which adapt efficiently and follow ethical guidelines [15]. Since the goal of the research is to enhance deepfake detection capabilities in a variety of ways, it concentrates on physiological signals and forensic techniques that scale. Our approach collects fixed-length video segments with facial portions, identifies regions of interest, and extracts biological signals from those segments. In the first section of our research, we analyse the pairwise separation issue, where video pairs are provided but the false one is unknown. We propose a solution by analysing collected biological signals, their conversions to various domains (time, frequency, time-frequency), and correlations. In the second section, we integrate findings from paired context and interpretable feature extractors in the literature.

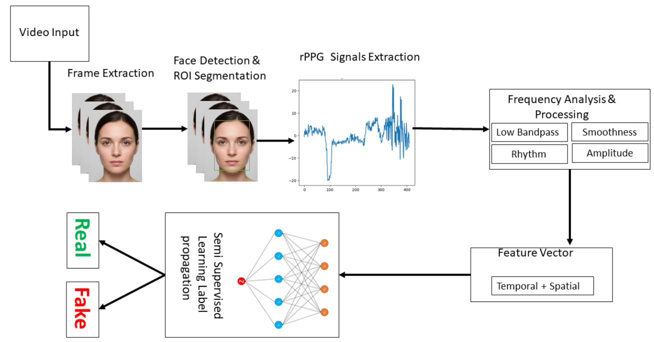

Fig. 1. The main methodology of forensic lense.

II. Literature Review

The paper ViGText: Deepfake Image Detection with Vision-Language Model Explanations and Graph Neural Networks, by Ahmad Albarqawi, Mahmoud Nazzal, Issa Khalil, Abdallah

Khreishah and NhatHai Phan (2025), puts forward a novel dual-graph detection model which combines image patches and textual explanations generated by a vision-language model (VLLM) into a single Graph Neural Network (GNN). This architecture detects visual and contextual discontinuities by incorporating spatial and frequency data representing patches of an image and matching them to patch-specific textual descriptions. Measured on datasets such as Stable Diffusion (SD), StyleCLIP, and adversarial image sets, ViGText achieves impressive results: under the generalization evaluation, accurate performance increases to 98.32 % / 99.21 % and 99.52 % / 99.60 % respectively; under the SD/StyleCLIP evaluation, it reaches 99.25 % / 99.90 % accuracy, 99.80 % / 99.90 % precision, 98.52 % ViGText achieves robust recall with under 11.1 % point worse than the baseline in foundation model-based adversarial attacks, and performance decreases under 4 % in stronger surrogate-based attacks. The model is practical (processes each image in approximately 1.76 seconds) despite its rich dual-graph architecture, which incurs a 0.10-second overhead relative to base methods that reflects its high balance between accuracy, generalizability, robustness, and efficiency. The only limitation in his work is that it is only applicable to still / static images and to be implemented on video, audio, and live streams we need to adjust it and enhance it. [3] Abdullah Alharbi et al. propose FDINet59, a 59-layer Fake Dense Inception Network, in their July 2025 article in Scientific Reports that is adapted to deepfake media-identification (specifically, media that circulates on social media). The model is trained on synthetic content created through autoencoders and GANs using faces that MTCNN has cropped. On training data, FDINet59 can get 70.02 % accuracy (log loss = 0.688), whereas evaluation data can achieve much higher 94.95 % accuracy with 0.205 log loss. Limitations are not explicitly described in the paper, leaving generalization, computing efficiency, and resilience to a variety of or novel deepfake methods open to question. [4] In 2024, a novel audio-visual deepfake detection model was suggested by Gao et al in their Electronics study that worked to predict temporal features by combining two streams of architecture design. The model uses pairs of adjacent audio-video segments as inputs (running visual data through a Channel-Separated 3D Convolutional Network (CSN) and audio through Mel-Frequency Cepstral Coefficients (MFCCs) into a ResNet-18 backbone) to make predictions of future features in each modality, with contrastive learning to align audio-visual embeddings across modalities. The model is tested with the FakeAVCeleb dataset, which consists of various types of forgery (fake video/fake audio, fake video/real audio, real video/fake audio, real video/real audio), resulting in an accuracy of 84.33 and an AUC of 89.91, even higher than both unimodal baselines (only visual (ACC 79.87 % AUC 80.54 %) and only audio (ACC 71.35 % AUC 72.26 %)). Its primary strength is that it combines both intra-modal temporal inconsistency modelling with inter-modal contrastive alignment which allows it to perform robust detection on complex multimodal manipulations. However, the lack of real-time testing and a comparatively simplistic audio processing pipeline (using just the MFCC features) raise concerns about future problems with application in actual streaming scenarios. [16] In their 2022 paper "Analysis of Score-Level Fusion Rules for Deepfake Detection", Sara Concas, Simone Maurizio La Cava, Giulia Orrù, Carlo Cuccu, Jie Gao, Xiaoyi Feng, Gian Luca Marcialis, and Fabio Roli explore improving generalization in deepfake detection via ensemble-based score-level fusion methods. They evaluate multiple base classifiers—namely, a ResNet50 visual artifact-based detector, XceptionNet general network-based detector, and four EfficientNetB4 variants (standard, Siamese-trained, attention-enhanced, and attention + Siamese)—using the FaceForensics++ dataset for training and intra-dataset testing and the DFDC dataset for cross-dataset evaluation. They test three fusion categories: non-parametric rules (average, Bayesian, product, max, and min), weighted-average (parametric) based on accuracy, correlation, or mutual information, and classification-model-based fusion (linear perceptron, SVM with RBF, multilayer perceptron, and Complement Naive Bayes). Results show that in the intra-dataset scenario, the best single model (EfficientNetB4ST) achieves an AUC around 0.959, while the MLP fusion rule yields an improved AUC of 0.984 with a lower equal-error rate (EER is approximate to 0.053) . In the more challenging cross-dataset scenario, Weighted-Average fusion using correlation-based averaging and the Complement Naive Bayes model both outperform single models, demonstrating superior generalization. The authors note that non-parametric fusion fails to consistently improve generalization, whereas parametric approaches (especially correlation-based weighted average and model-based fusion) provide robust gains, though at the cost of offline weight estimation. Limitations highlighted include dependency on validation data distributions (which may introduce bias), the offline tuning burden of parametric fusion, and the need for broader evaluation using models trained on diverse datasets and fusion strategies (e.g., decision-level or unsupervised fusion) [12] Aminollah Khormali and Jiann-Shuang Yuan (2021) suggest a flexible enhancement architecture called ADD (Attention-based DeepFake Detection), which is meant to complement the existing CNN classifiers (e.g., VGGNet, ResNet, Xception, MobileNet), to detect video deepfakes. ADD applies face-focused data augmentation, such as face close-up and face shut-off, to highlight forged parts, and then applies attention-based supervision to compel the model target those localized forgery regions during learning. Measured on the Celeb-DF v2 and WildDeepFake datasets, ADD can significantly improve detection: ADD-ResNet obtains more than 98.3 % AUC in Celeb-DF v2, a large improvement over the baseline models. Its main advantage is to amplify generalization and increase detection accuracy of various CNN backbones through region-based attention. However, the study does not provide cross-dataset robustness measurements, resource/performance overhead, or other architecture results, which leaves unresolved the problems of computational efficiency and broader application. [30] Lukas Kroiss and Johannes Reschke present a simplified but very precise ResNet-50-based CNN classifier fine-tuned to detect

TABLE I
SUMMARY OF VISUAL BASED DEEPFAKE DETECTION TECHNIQUES

Year/ paper	**Technique**	**Methodology**	**Dataset Used**	**Results / Accuracy**	**Strengths**	**Limitations**
2025 [3]	ViGText	Dual-graph detection (image patches + VLLM text explanations into GNN)	Stable Diffusion, StyleCLIP, adversarial sets	Accuracy/F1 up to 98.0%;	Captures visual + contextual cues; robust against adversarial attacks; efficient (1.76s/image)	Limited to static images; no video/audio/real-time support
2025 [4]	FDINet59	59-layer Dense Inception CNN trained on cropped faces (MTCNN)	Synthetic autoencoder + GAN generated content	Training Acc: 70.02%, Eval Acc: 94.95% (log loss 0.205)	Tailored for social media; achieves high evaluation accuracy	No explicit limitations; unclear generalizability, robustness, or efficiency
2025 [33]	ResNet-50 Classifier	Fine-tuned ResNet-50 with sigmoid dense layer for binary classification	Diverse Face Fake Dataset (DFFD)	AUC 89.91%	Combines transfer learning with diverse dataset; strong single-image detection	Not tested on videos; lacks cross-domain evaluation
2025 [17]	Time-Aware Fine-Tuning	Timestamp-conditioned tuning module for adaptability	FF++, FF++-C40, FF++ (2025)	84.4% Acc, 0.982 AUC (vs. base 74.2%, 0.935 AUC)	Adaptable to evolving deepfake generation methods; forward-compatible	Relies on accurate timestamps; weaker in cross-domain or unlabeled data
2024 [16]	Audio–Visual Framework	Dual-stream model with CSN (video) + MFCC + ResNet-18 (audio); contrastive audio-visual alignment	FakeAVCeleb	84.33% Acc, 89.91% AUC; unimodal video 79.87%, audio 71.35%	Strong multimodal fusion; handles complex manipulations well	Audio pipeline limited (MFCC only); no real-time evaluation
2024 [46]	ResNet-Swish-BiLSTM	Hybrid CNN-LSTM (ResNet backbone + Swish activation + BiLSTM)	FaceForensics++, DFDC	96.23% Acc (FF++), 78.33% Acc (FF++ + DFDC)	Captures temporal dependencies across frames; robust in forensic scenarios	Lower cross-dataset accuracy; dataset bias sensitivity
2024 [10]	PPG-Based Source Detection	Heartbeat-induced facial PPG signals with classification network	Multiple datasets (portrait videos)	97.29% deception detection, 93.39% source model ID	High robustness across datasets and demographics; dual-purpose (real/fake + source)	Dependent on strong physiological signals; weaker on low-quality videos
2022 [12]	Fusion Rules	Ensemble-based score-level fusion (ResNet50, Xception, EfficientNetB4 variants); non-parametric, weighted-average, and model-based rules	FaceForensics++, DFDC	Intra-dataset: Best single AUC 0.959; Fusion (MLP) AUC 0.984 (EER 0.053); Cross-dataset: WA & CNB outperform single models	Improves generalization with parametric/model-based fusion; effective cross-dataset robustness	Non-parametric weak; parametric requires offline tuning; validation bias risk; limited evaluation scope
2022 [23]	Convolutional Attention Network (CAN) with rPPG signals	Celeb-DF v2, DFDC	>98% AUC (single-frame); 100% Acc (Celeb-DF v2 with fusion)	Detects physiological inconsistencies invisible to humans; robust against visual artifacts	Sensitive to video quality, compression, and resolution (affects rPPG signals)	Dependent on strong physiological signals; weaker on low-quality
2021 [30]	ADD	Attention-based enhancement with face-focused augmentation + attention-driven supervision	Celeb-DF v2, WildDeepFake	ADD-ResNet AUC 98.3%+ on Celeb-DF v2 (beats baselines)	Boosts generalization; works across CNN backbones; region-specific attention improves accuracy	No cross-dataset robustness results; lacks computational overhead analysis; not applied to all CNNs

single face images as deepfakes in their 2025. By using a large-scale so-called Diverse Face Fake Dataset (DFFD), consisting of various types of manipulations, including DeepFakes, Face2Face, FaceSwap and NeuralTextures with different identities, they made the final model by replacing the top-layer of ResNet-50 with a sigmoid-activated dense layer to classify authenticity. As shown, the model has high performance in terms of detection, with a precision of 0.98%, recall of 0.96%, F1-score of 0.97%, and AUC of 0.99%, indicating its ability to simultaneously split fake and authentic images in diverse settings. Its major advantage is its ability to effectively transfer learning to a heterogeneous dataset allowing robust, high-fidelity single-image deepfake detection. Its lack of evaluation on films or cross-domain datasets, which might test generalizability to compression or invisible manipulation forms, is one potential limitation. [33] Jiaquan Zhang et al. suggest a new time-aware fine-tuning approach in their 2025 article Till This Very Moment: Timestamping the Latest Deepfake Detection Model via Time-Aware Fine-Tuning to adjust existing deepfake detectors to be useful against those generated by newer generation models. They test their framework, using synthetic content as input, on the FF++, FF++-C40, and a newly introduced FF++ (2025) dataset of content that was created in late 2024 by configuring a timestamp-conditioned tuning module to modify the model behavior according to the creation date of the synthetic content. The fine-tuned model is shown to be resilient as it has high detection metrics- 84.4 % accuracy and 0.982% AUC on FF++ (2025) compared to the original base model of 74.2 % accuracy and 0.935% AUC. Its time-sensitive flexibility, which offers a future-proof approach that manages future deepfake complexity, is its most significant asset. However, when using poorly described or unlabeled data, its reliance on exact timestamp metadata and poor performance in cross-domain use cases suggest flaws in real-time adaptability and robustness. [20] Abdul Qadir et al. suggest the hybrid deep-learning model, ResNet-Swish-BiLSTM, a combination of a backbone of ResNet and Swish activation and BiLSTM in their 2024 research article to identify deepfake videos by examining consecutive targeted frames. They test the model using FaceForensics++ (FF++) and Deepfake Detection Challenge (DFDC) datasets and report 96.23% accuracy on the FF++ dataset and 78.33% accuracy on the FF++ and DFDC data, showing that the model is robust to mixed deepfake sources.The model's ability to replicate temporal dependencies and artifacts between frames makes it suitable for use in real-time forensic scenarios, which is one of its main advantages. However, the lower cross-dataset testing performance suggests that generalizability is weak, especially when dealing with heterogeneous data sources, and that it might be susceptible to bias in the dataset and other manipulation techniques. [46] The authors of Deepfake Source Detection in a Heart Beat (Ciftci et al., 2024) offer both a single-purpose and dual-purpose detection framework, which, firstly, can identify genuine and fake portrait videos but, secondly, can also define what generative model has created the fake. This technique uses PPG cell analysis, which records color changes in the face caused by changes in heartbeat and is then subjected to an advanced classification network. The technique can be considered highly robust to a wide range of datasets, demographic settings, and post-processing perturbations, with 97.29% accuracy in detecting deception and 93.39% accuracy in identifying the source model. However, because it relies on powerful physiological signs, its effectiveness may be compromised by low-quality or severely distorted inputs, and the technique's competency with different generative model types may be crucial to accurately identifying the source. [11] In DeepFakesON-Phys (2022), Javier Hernandez-Ortega, Ruben Tolosana, Julian Fierrez, and Aythami Morales present a Convolutional Attention Network (CAN) that uses physiological artifacts to detect deepfakes on face video frames by using remote photoplethysmography (rPPG) to estimate heart rate information. This model is tested against the Celeb-DF v2 and DFDC benchmark datasets, and on both datasets, the model scores above 98% in AUC in single-frame analysis. Besides, with continuous frame-score fusion using heuristic and statistical methods, the system achieved 100 % accuracy on Celeb-DF v2 at a low latency. The key advantages of the method are that it is sensitive to invisible physiological anomalies in the visual artifact, and it is also resistant to standard manipulation. Its dependency on the quality of physiological signals may however restrict use in the real world where subtle rPPG signals are obscured by video compression, noise, or low resolution. [23] Wang et al. presented VCapAV, a 252-hour dataset used to identify audio-visual deepfakes, at Interspeech 2025. This dataset takes into account background sound alterations in addition to speech.The dataset is a mixture of real and fake audio (created with AudioLDM, Audiocraft, V2A models) and fake video (Kling), which is filtered using captioning and multi-stage validation to come up with a total of approximately 91k clips.The benchmarks showed that whereas ResNet18 and LightCNN struggled with generalization (EER 14%), AASIST achieved up to 99.9% accuracy on observed manipulations and 96% accuracy on unknown manipulations.Meso4 also achieved lower accuracy of 54.5 % on video-only detection, compared to 75 % on other datasets, which demonstrates that VCapAV is more severe. Weaknesses are the fact that the number of fake video samples is small and that it is not the full multimodal fusion. The article also emphasizes the importance of the strong multi-mod detectors against various forms of non-speech deepfakes. [50] Muruganandham et al. (2025) developed LSTM-AE-DRDE, a deepfake audio detection framework that combines a dynamic residual encoding lossless block with an attention-enhanced LSTM autoencoder.The model delivers high detection rates with different audio features such as MFCC, wavelet, prosodic, temporal, and glottal features as well; in-dataset (85-97) and cross-dataset (up to 95) generalization (e.g., CMFD) with aggressive EER and ROC-AUC values. It outperforms the conventional deep learning models and matches SOTA results on the ASVspoof 2021 benchmark. Although it is very good in performance, additional testing and analysis in multilingual as well as real-life and resource-constrained environment will aid in measuring if it has a wider usage. [43] Kevin Warren, Daniel Olszewski, Seth Layton, Kevin Butler, Carrie Gates, and Patrick Traynor suggested a new type of detection based on acoustic prosodic features, i.e. pitch, jitter, shimmer, harmonic-noise ratio (HNR), and intonation to distinguish between deepfake audio and natural speech in their preprint, Pitch Imperfect: Detecting Audio Deepfakes with Acoustic Prosodic Analysis, dated February 2025. They use their model with six standard prosodic features on the ASVspoof2021 dataset and give a detection rate of 93 % with an Equal Error Rate (EER) of 24.7 %. The approach has a high explainability with Jitter, Shimmer, and Mean Fundamental Frequency as the most imperative features and it is also resistant to adversarial attacks, even with infinity-norm attacks, the model remains resilient whereas baseline models lose accuracy with up to 99.3 % accuracy. One of the main strengths of this method is its ability to be interpreted and use human-perceivable speech cues,, strengthening both detectability and resiliency. The drawbacks are however, relatively high EER which suggests an improvement and potential sensitivity to

TABLE II
SUMMARY OF BIOLOGICAL SIGNAL/RPPG-BASED DEEPFAKE DETECTION

Year/paper	Technique	Methodology	Dataset Used	Results / Accuracy	Strengths	Limitations
2025 [33]	ResNet-50 Classifier	Fine-tuned ResNet-50 CNN with sigmoid dense layer for binary classification of single face images	Diverse Face Fake Dataset (DFFD) containing DeepFakes, Face2Face, FaceSwap, NeuralTextures	Precision 0.98%, Recall 0.96%, F1-score 0.97%, AUC 0.99%	Strong transfer learning; robust single-image detection across manipulations	Not tested on videos or cross-domain datasets; limited generalizability under compression or subtle manipulations
2025 [17]	Time-Aware Fine-Tuning	Timestamp-conditioned tuning module adapting deepfake detectors to new-generation fakes	FF++, FF++-C40, FF++ (2025) synthetic dataset	84.4% Acc, 0.982% AUC (vs. base 74.2%, 0.935% AUC)	Future-proof adaptability; resilient against evolving deepfake techniques	Relies on precise timestamps; weak cross-domain robustness and performance on unlabeled data
2024 [46]	ResNet-Swish-BiLSTM	Hybrid CNN-LSTM with ResNet backbone, Swish activation, and BiLSTM for temporal frame analysis	FaceForensics++ (FF++), Deepfake Detection Challenge (DFDC)	96.23% Acc (FF++), 78.33% Acc (FF++ + DFDC)	Captures temporal dependencies; strong in forensic and real-time scenarios	Lower cross-dataset performance; dataset bias; weak generalization on heterogeneous sources
2024 [11]	Deepfake Source Detection in a Heart Beat	PPG-based dual-purpose model using heartbeat-induced facial color variations and classification network	Multiple portrait video datasets	97.29% deception detection; 93.39% source identification accuracy	Robust across demographics and datasets; detects both authenticity and generative source	Depends on strong physiological signals; less effective on low-quality or distorted inputs
2022 [23]	DeepFakesON-Phys	Convolutional Attention Network (CAN) leveraging rPPG signals for physiological anomaly detection	Celeb-DF v2, DFDC	>98% AUC (single-frame), 100% Acc (Celeb-DF v2 with fusion)	Detects invisible physiological inconsistencies; robust to visual artifacts	Sensitive to compression, noise, and low-resolution videos affecting rPPG quality

audio quality e.g. compression artifacts or background noise which can impair the fidelity of prosodic features. [58] The authors suggest using a multimodal emotion-aware deepfake detector, which integrates physiological (e.g., remote photoplethysmography - rPPG) and behavioral (facial expressions, speech prosody, and micro-expressions) signals in their study in 2024. The rationale is that existing methods of deepfaking typically generate surface-grading realism, and do not recreate emotion-physiology consistency, including heart rate variations in line with frailty reactions via facial or vocal manifestations. The system combines vision transformers of physiological signals estimation and multi-branch deep networks of behavioral emotion features and fuses them through a fusion strategy that takes advantage of attention mechanisms. Findings indicate that the fusion model is much better than the unimodal baselines (video-only or audio-only) to achieve up to 94.2% accuracy and 92.8% F1-score on multimodal benchmark datasets. The innovative integration of emotion-physiology consistency checks can be considered as a strength and bring an understandable aspect to deepfake detection.Its higher processing cost and reliance on high-quality input signals are drawbacks, which may reduce robustness in low-resolution, noisy, or compressed environments. [27] The paper by Yipin Zhou and Ser-Nam Lim (2021) of ICCV suggests a two-way joint audio-visual deepfake detector leaning on the inherent synchronization between the video and audio streams-by introducing a joint audio-video system known as a sync-stream network architecture, which combines both modalities with attention to enhance detection. On FaceForensics++ (FF) and DFDC video sets, the sync-stream model provides an 99.19% video-level user accuracy and 77.85% audio-level user accuracy, leading to 87.40 % accuracy when both streams are viewed in concert. Attention also enhances performance: sync-stream attention gives 99.99% (video), 84.66% (audio), and 94.36% (joint) FF, 90.48% (video), 96.01% (audio), and 89.39% (joint) DFDC. Such outcomes prove that the intermodal synchronization modeling helps establish a large improvement in the detection performance and the robustness. Nevertheless, the mechanism is based on synchronized and uncorrupted audio-visual information and adds new architectural complexity in terms of attention mechanisms–other elements that can complicate implementation in real-life noisy or imbalanced audio-visual feedback. [62] Mahmudul Hasan (May 2025) describes a deep learning-based framework that implements MTCNN to detect the faces and EfficientNet-B5 to act as an encoder to detect deepfake videos with the use of the Kaggle DFDC dataset. The model has a log loss of 42.78%, an AUC of 93.80% and an F1 of 86.82% indicating excellent detection. Its strengths are the use of strong facial preprocessing and strong CNN backbone that provides strong classification measures that can be applied practically. Nevertheless, the absence of cross-dataset testing, real-world video analyses, and comprehensive robustness analysis point to the principal limitations in the evaluation of the real-world generalizability

TABLE III
SUMMARY OF AUDIO BASED DEEPFAKE DETECTION

Year	Technique	Methodology	Dataset Used	Results / Accuracy	Strengths	Limitations
2025 [57]	VCapAV	252-hour dataset for audio-visual deepfake identification; includes speech and background sound alterations; validated via captioning and multi-stage filtering	Real/fake audio (AudioLDM, Audiocraft, V2A) + fake video (Kling); 91k clips	AASIST:99.9% (known),96% (unknown); ResNet18/LightCNN EER >14%; Meso4:54.5% (video-only), 75% (others)	Comprehensive multimodal dataset; highlights robustness of multi-modal detectors against non-speech deepfakes	Few fake video samples; lacks complete multi-modal fusion; dataset imbalance
2025 [43]	LSTM-AE-DRDE	LSTM autoencoder with dynamic residual encoding and attention-enhanced feature learning for audio deepfake detection	ASVspoof 2021, CMFD, other audio benchmarks	In-dataset 85–97%, cross-dataset up to 95%; strong EER and ROC-AUC	High generalization; robust across multiple audio feature types (MFCC, wavelet, prosodic, temporal, glottal)	Needs testing in multilingual and real-world noisy conditions; unknown efficiency in resource-constrained systems
2025 [58]	Pitch Imperfect	Acoustic prosodic feature-based detection using pitch, jitter, shimmer, HNR, and intonation for explainable fake speech analysis	ASVspoof 2021	93% detection rate; 24.7% EER; robust under adversarial (-norm) attacks with up to 99.3% accuracy	High interpretability; uses human-perceivable cues; resilient to adversarial attacks	High EER; sensitive to compression artifacts, background noise, and low-quality audio
2021 [62]	Sync-Stream Network	Joint audio–visual deepfake detector leveraging synchronization attention between modalities	FaceForensics++ (FF), DFDC	FF:99.99% (video), 84.66% (audio), 94.36% (joint); DFDC: 90.48% (video), 96.01% (audio), 89.39% (joint)	Strong intermodal synchronization; large performance boost from attention; improved robustness	Requires synchronized, clean AV input; complex architecture; harder real-world deployment

and resilience. [2] In their 2024 Electronics paper, C.Y. Lin and colleagues suggested a spatiotemporal deepfake detector that is specifically tailored to identity-swap video manipulations. The model combines facial landmark analysis, which follows 68 salient locations, with attention-directed data augmentation (AGDA) to emphasize manipulation-evident areas, at the same time, a combination of spatial features and temporal facial integrity are achieved. Their method has much higher levels of preciseness in comparison to rival models (FakeVideoForensics, DeepFakes-FacialRegions, Improved Xception, AltFreezing) with a recognition of 98.13%, 97.94%, 97.87% and 98.6198.61% all being achieved in the four datasets evaluated-UADFV, FaceForensics++, Celeb-DF and DFDC respectively, and the average of 98.14% is among 98.13%, 97.94%, The effectiveness of this method is in the balanced use of time and space information, which increases the level of resistance to various manipulations of videos. Nevertheless, more specific measures like AUC or inference latency are not given, and generalization of the model to unknown manipulation methods or identity swaps scales is yet to be done. [37] G. Naskar et al. (2024) also suggest a stacking-based ensemble deepfake detector, which combines the results of two state-of-the-art CNNs–Xception and EfficientNet-B7–after which the results are refined by a feature selection mechanism based on ranking and finally classifies the results with the help of a meta-learner (Multi-layer perceptron) in their open-access paper. The model is assessed on two major video deepfake datasets, Celeb-DF (v2) and FaceForensics, with the model getting 96.33% accuracy on Celeb-DF (v2) and 98.00% accuracy on FaceForensics. The main strengths of this approach are its capacity to achieve better results than the base models by feature-level fusion and meta-learning, a physical realization of robustness in various techniques of deepfake generation and perturbations. The use of stacking and feature ranking however add more complexity and computation cost to the model which may limit real time deployments and scale, [44] The author, in Deepfake Detection Using the Rate of Change between Computer Vision Features (Lee, 2021), presents a scale- and memory-efficient deepfake detector based on the temporal variations of the traditional computer vision features, instead of using CNN-based models only. The process has been used to extract features of MSE, PSNR, SSIM, color histograms, edge density, and DCT coefficients, and their frame-to-frame change is modeled using a deep neural network (DNN). The subsets of FaceForensics++ (Face2Face, FaceSwap) and DFDC datasets were experimented on with 300 frames per video being preprocessed using MTCNN. The proposed model was found to reach 95.22% accuracy when the DNN was running on variance over 20 frame window and 97.39% accuracy when Adam optimizer and five hidden layers were used. It shows that temporal variability is a computationally efficient method to substitute CNN-heavy networks, with fewer parameters (approximately 15k vs. approximately 28k) and loss of 30 % of training time. The framework also was hardly susceptible to distortions like blur, noise, and changes in brightness. Although, the main limitation is that it relies on various sequential frames thus less efficient in the analysis of single images or data with unevenly distributed frames. [13] In the article Deepfake Video Detection Using Convolutional Vision Transformer (2021), the authors propose a hybrid model that uses Convolutional Neural Networks (CNNs) and Vision Transformers (ViT) to detect video-based deepfakes. The CNN backbone extracts local facial features (i.e. texture, edges, micro-expression) first before sending it into ViT encoder to help identify long-range spatial dependencies and global contextual cues across the frames. The model's two-stage structure allows it to take advantage of the trade-offs between more broad structural inconsistencies and fine-grained pixel-level representations, which are readily missed by single-stage CNN-based techniques. The system was trained and tested on DeepFake Detection Challenge (DFDC) dataset which was a large scale benchmark of millions of manipulated and genuine videos. The experimental findings indicate that the proposed architecture attains the detection accuracy of 91.5% and AUC of 0.91%, which is superior to the traditional CNN-only baselines. The authors draw a conclusion that convolutional feature extraction with transformer-based attention is much more resistant to attack deepfake detection, particularly in the context of real-world distortions (compression and noise). [60] Misaj Sharafudeen and Vinod Chandra S S present a framework of frequency forensics to detect deepfakes using face on the example of a Dual Residual Network (DRN) in their 2025 article Frequency Forensics for Deep Fake Face Detection Using Dual Residual Networks, where frequency-domain forensic evidence is used to forecast deepfakes. The model produces surprisingly low Equal Error Rates of 0.04% on DFFD PGGAN, and 0.02% on both DFFD StyleGAN and the Stable Diffusion part of the DFF dataset, by removing residual traces of high-frequency components of facial images, which are generated by PGGAN, StyleGAN, and Stable Diffusion, and comparing them to real images via Representation Similarity Analysis (RSA). This architecture is very efficient in contrasting between synthetic and original images, but the paper indicates a small increase in AUC of 40.89% of Stable Diffusion content. This implies that the DRN is incredibly accurate, but with limited details of performance on certain generative styles. Its biggest strength is that it is able to identify faint high-frequency residual artifacts on a scale unmatched by other methods in a wide range of deepfake generators. Nonetheless, the extent of its extrapolation of these face-generation techniques and post-processing (e.g., compression or blurring) resistance is uninvestigated and should be reaffirmed. [52] Maryam Abbasi, Paulo Vaz, Jose Silva, and Pedro Martins in their article on Applied Sciences of 2025 thoroughly test three convolutional neural network models -Xception, ResNet-50 and VGG16-based convolutional neural networks- to identify frame subsets in DFDC and FaceForensics++ datasets as deepfakes. They are analyzed by metrics such as accuracy, precision, F1-score, AUC-ROC, and resilience to adversarial conditions (through FGSM attacks) in their analysis. Xception is the most accurate model, reaching 89.2% accuracy on DFDC and 85.7% on FaceForensics++, and as the most suitable option, it has strong generalization and is fast to inference with a throughput of around 85 ms per frame.VGG16 has a lower inference speed (around 1020 ms/frame), but it is competitive in terms of precision and recall (F1-score of about 87.0%). ResNet-50 has a lower AUC on adversarial example perturbations and a weaker generalization, but it can be trained considerably more quickly (currently at 270 ms/frame) and more readily (DFDC: 72.8).Even though Xception is unique in terms of deployed to real-life settings, as it is fast and at the same time, more accurate, it is important to note that any model has a significant drop in performance in adversarial settings, which would result in more robust and resilient detection systems. [38] Wasin Alkishri, Setyawan Widyarto, and Jabar H. Yousif (2024) in their Journal of Internet Services and Information Security article examine whether it is possible to remove GAN fingerprints of synthetic images to deceive deepfake detectors. They test the effect the removal of high-frequency GAN artifacts in StyleGAN-generated images in a 140K real-and-fake face dataset using frequency-domain analysis and discrete fourier transforms, and analyze the effect on detection using the XceptionNet model. Following fingerprint removal, 99.78% of manipulated images were tagged as true and the accuracy, precision, recall and F1-score and AUC were approximately 99.32% and showed that it was almost in its entirety deceptive. However, on real and fake images tested on a 50% real and an equal amount fingerprint-removed image set, detection dropped to chance-accuracy and AUC reduced to about 50% demonstrating the failure of the detector to differentiate between real and concealed-fingerprint images. This points out the susceptibility of the existing deepfake detection methods to basic frequency-based defenses. However, the authors mention that they only consider StyleGAN images and may not generalize to other types of GAN architectures or perturbations, and suggest that further assessment should be done on a variety of data sets and attacks. [14] The authors in Deep fake detection using cascaded deep sparse auto-encoder for effective feature selection (Balasubramanian et al., 2022) offer a new detection model that incorporates a Cascaded Deep Sparse Autoencoder (CDSAE) trained using a Temporal Convolutional Neural Network (TCNN), which extracts frame-level visual features and then classifies them using a Deep Neural Network (DNN). The method was tested on deepfake video benchmarks, like Face2Face, FaceSwap, and DFDC: the detection accuracy of the method was 98.7%, 98.5%, and 97.6%, respectively, much higher than baseline models, including ResNet, MobileNet, and classic SVM. Also, the method showed a reduced processing time and increased scores of AUC. The paper proposes an excellent direction of both increasing the effectiveness and reliability of the deepfake detection systems by highlighting the use of unsupervised feature extraction, which is more attuned to the temporal consistency and decreasing the overfitting with the use of sparse autoencoding. [8] Li et al. (2022) report a one-class detection model in their sensors article, which differentiates GAN-generated face images, based on a new Multi-Channel

TABLE IV
SUMMARY OF VIDEO-DYNAMICS DEEPFAKE DETECTION TECHNIQUES

Year	Technique	Methodology	Dataset Used	Results / Accuracy	Strengths	Limitations
2025 [22]	EfficientNet-B5 + MTCNN	MTCNN for face detection; EfficientNet-B5 encoder for deepfake video classification	Kaggle DeepFake Detection Challenge (DFDC)	Log loss 42.78%, AUC 93.80%, F1 86.82%	Strong facial preprocessing; powerful CNN backbone; practical classification results	No cross-dataset or real-world testing; lacks robustness evaluation; unclear generalizability
2024 [37]	Spatiotemporal Identity-Swap Detector	Combines facial landmark tracking (68 points) with attention-guided data augmentation (AGDA) for spatial-temporal feature fusion	UADFV, FaceForensics++, Celeb-DF, DFDC	UADFV 98.13%, FF++ 97.94%, Celeb-DF 97.87%, DFDC 98.61% (avg. 98.14%)	Balanced use of spatial and temporal cues; strong resistance to varied manipulations	Missing AUC and latency data; limited testing on unseen manipulation types
2024 [44]	Stacking-Based Ensemble	Ensemble of Xception + EfficientNet-B7 with feature selection and meta-learning (MLP classifier)	Celeb-DF (v2), FaceForensics	96.33% (Celeb-DF v2), 98.00% (FaceForensics)	Outperforms base models; robust feature-level fusion; effective across multiple manipulations	High complexity and computational cost; less suitable for real-time deployment
2021 [34]	Temporal Feature Change Detector	DNN models temporal variations of traditional vision features (MSE, PSNR, SSIM, color, edge, DCT) between frames	FaceForensics++ (Face2Face, FaceSwap), DFDC	95.22% Acc (20-frame window), 97.39% Acc (optimized model)	Lightweight; faster training; robust to blur, noise, and brightness distortions	Needs sequential frames; limited single-image efficiency
2021 [26]	CNN–ViT Hybrid (Deepfake Video Detection Using Convolutional Vision Transformer)	Two-stage hybrid combining CNN feature extraction and ViT for long-range spatial dependencies across frames	DeepFake Detection Challenge (DFDC)	91.5% Accuracy, 0.91% AUC	Captures both local texture and global context; resistant to compression/noise attacks	Requires high computational resources; lacks temporal modeling

Convolutional Neural Network (MCCNN), which includes two steps of training with attention-guided weakly supervised learning and data augmentation.It is tuned to detect unknown GAN methods using a one-class classification loss and trained to detect the existence of known false features using binary classification loss.The filters that are applied during data augmentation include filtered enhancements (Gaussian blur, noise, motion blur, homomorphic filtering, Fourier spectrum) as well as attention cropping and dropping. The model is evaluated on a ProGAN (source-domain) vs. StyleGAN, StyleGAN2, BigGAN, DCGAN, DeepFake, and VQ-VAE2.0 (cross-domain) configuration and is found to be of very high quality: its source-domain accuracy is 99.4% for real images and 98.9% for ProGAN fakes, with the F1-score approximate to 0.992%.This is an accuracy improvement of 5-30% and dramatic improvement in F1-score over previous methods. Its key advantages are improved generalizability between unseen GAN models, the effective utilization of attention-enhanced one-class models, and good source-domain results. Nevertheless, there are still constraints (performance drops considerably with much different types of GANs, such as 80.9% in VQ-VAE2.0) as well as sensitivity to the heterogeneity of augmentation techniques and training settings. [35] In the article "Audio Deepfake Detection Using Deep Learning" (Shaaban and Yildirim, 2025), the researchers present a new model of the StacLoss, a specific contrastive loss function, and self-attention modules to improve differentiation between genuine and fake audio samples Research,GateDOAJ. The architecture employs layered multi-head attention to extract rich, discriminative features from raw audio pairs after passing them through two convolutional branches connected by residual links.StacLoss (reduces the distance between authentic audio samples of identical identity), and maximizes distance between manipulated ones- amplifies the capacity of the model to detect subtle fakes ResearchGateDOAJ. Tested on the benchmark ASVspoof2019 data, the model reported a high performance with an accuracy of 98%, precision of 97%, recall of 96%, F1 score of 96.5% with ROC-AUC of 99% and an EER of only 2.95%. [51] Javed et al. (2024) in the Electronics study suggest a real-time deepfake video detection system to be a hybrid of the lightweight MesoNet4 system (to detect manipulations on the face that are subtle) and the feature-rich ResNet-101 (to represent complex visual images) deep learning-based system. The hybrid model is evaluated on FaceForensics++, CelebV1 and CelebV2 datasets and the results are impressive: 98.73% on FaceForensics++, 96.89% on CelebV1 and 97.90% on CelebV2 demonstrating the good performance of the hybrid model in terms of its generalizability and resilience in video forensics use. The combination of eye movement cues and dual-model feature extraction is its main advantage that increases detection accuracy during live-streams. Nevertheless, the existing performance is strong in a variety of benchmark datasets, but it has drawbacks such as the ability to change depending on the light variation and its computational efficiency in the context of being used in the actual live-stream scenario, including the issues of real-time processing on consumer-level

TABLE V
SUMMARY OF FREQUENCY DOMAIN BASED DEEPFAKE DETECTION TECHNIQUES

Year	Technique	Methodology	Dataset Used	Results / Accuracy	Strengths	Limitations
2025 [52]	Frequency Forensics for Deep Fake Face Detection Using Dual Residual Networks (DRN)	Leverages frequency-domain forensic evidence via Dual Residual Networks (DRN) to identify fake faces.	DFFD PGGAN, DFFD StyleGAN, DFF (Stable Diffusion subset)	EER: 0.04% (PGGAN), 0.02% (StyleGAN, Stable Diffusion); AUC +40.89% (Stable Diffusion)	Extremely accurate in detecting faint high-frequency residual artifacts across diverse generators.	Limited robustness testing for compression, blur, or unseen generative styles.
2024 [5]	Frequency-Domain GAN Fingerprint Removal Study	Explores frequency-domain effects of GAN fingerprint removal on detection using XceptionNet.	StyleGAN (140K face dataset)	Accuracy: 99.32%; AUC 50% post-fingerprint removal; F1 99.3% before removal	Highlights vulnerability of detectors to frequency-based concealment of GAN artifacts.	Focused solely on StyleGAN; lacks generalization to other GAN types or perturbations.
2022 [8]	Cascaded Deep Sparse Autoencoder (CDSAE)	Combines Temporal CNN and DNN classifier for temporal deepfake detection via unsupervised feature extraction.	Face2Face, FaceSwap, DFDC	Accuracy: 98.7%, 98.5%, 97.6%; improved AUC; faster processing than ResNet, MobileNet, SVM baselines	Efficient unsupervised learning; enhances temporal consistency and minimizes overfitting.	Limited validation on large cross-domain datasets; lacks adversarial manipulation testing.
2022 [35]	Multi-Channel CNN (MCCNN)	Attention-guided weak supervision with one-class classification loss for cross-domain fake detection.	ProGAN (source) vs. StyleGAN, StyleGAN2, BigGAN, DCGAN, DeepFake, VQ-VAE2	Accuracy: 99.4% (real), 98.9% (ProGAN); F1: 0.992%; Cross-domain: 80.9% (VQ-VAE2)	Strong generalization to unseen GANs; attention + one-class learning improve robustness.	Accuracy drops with distinct GAN families; sensitive to augmentation and parameter diversity.

hardware. [24] In a study published in Electronics, Awotunde et al. (2023) present a five-layer convolutional neural network (CNN) that is specifically designed for deepfake video detection and classification. Additionally, the network utilizes optimal ReLU activations to efficiently identify features inside the facial regions.The framework is tested on difficult data sets using DeepFake and First-Order Motion (Face2Face) manipulations and attains a high prediction rate of 98% and 95% respectively on Deepfake and Face2Face respectively at real network conditions. When directly compared to the benchmark models like Meso4, MesoInception4, Xception, EfficientNet-B0 and VGG16, their network performs the most overall- with an average accuracy of 86 % under a variety of conditions. The strength of the system is that it has a lightweight architecture, which is optimised to run in real-time with the high precision and effective computing. The paper however observes that although the performance on DeepFake datasets is excellent, the generalization to other manipulation techniques, video formats, or even compressed real-world streams has not been evaluated yet, which can be used to evaluate and expand the study. [7] The hybrid framework discussed in the article by Rimsha Rafique, Rahma Gantassi, Rashid Amin, Jaroslav Frnda, Aida Mustapha, and Asma Hassan Alshehri (2023) in the Journal of Scientific Reports is a hybrid system of deepfake images detection that uses Error Level Analysis (ELA) preprocessing and Convolutional Neural Network (CNN) feature extraction and classification with Support Vector Machines (SVM) and K-Nearest Neighbors (KNN). The method is evaluated on an image dataset (which is assumed to consist of both real and manipulated samples) and reaches a high accuracy of 89.5% when the features extracted by ResNet-18 are compared with an SVM classifier Nature. The major advantage of this model is the ability to use ELA to emphasize pixel-level manipulations, which are useful in supplementing CNN-extracted features to achieve successful classification. Its applicability, however, seems to be restricted to static images only; the authors admit that it would be necessary to apply the approach to video datasets and consider other architectures- which points out at the shortcoming of the generalizability and applicability of dynamic media. [48] In an article published in the Arabian Journal of Science and Engineering in 2022, Janavi Khochare, Chaitali Joshi, Bakul Yenarkar, Shraddha Suratkar and Faruk Kazi propose a two-modality framework of audio deepfake detection and compares a conventional feature-based method (where spectrograms are used as features) to an image-based one: audio is converted to mel-spectrograms and used to feed deep learning frameworks, i.e., Temporal Convolutional Networks (TCN) and With the Fake or Real (FoR) dataset, which is speech generated by an advanced text-to-speech system, they document an accuracy of 92% in the test of the TCN model, on the one hand, which is significantly higher than the classical machine learning techniques. This shows the ability of the model to use temporal correlations in audio to be effective in deepfake detection and its ability to maintain competitive accuracy to other CNN-based models such as VGG-16 and XceptionNet. One of the strengths is the ability to use the sequential modeling of features based on audio to ensure effective detection. Nonetheless, the analysis does not seem to consider the accuracy only without other measures, like AUC or EER, and does not compare it with different datasets and this is why it may be doubted that it can be generalized and is applicable in cross-domain settings. [29] In the paper Deep fake detection and classification using error-level analysis and deep learning (Rafique et al., 2023), the authors provide an automated system to detect and classify deep fake images. It starts with the application of Error Level Analysis (ELA) to show areas where an image can be altered.To create fine, deep features, these ELA outputs are then fed via convolutional neural networks (CNNs), such as ResNet-18 and GoogLeNet. The generated feature representations are then classified using either K-Nearest Neighbors (KNNs) or Optimized Support Vector Machines (SVMs).The method was tested on a mixed dataset, with the highest accuracy of 89.5% with the features of ResNet-18, and a KNN classifier. This evidences the strength and success of forensic preprocessing real-time deepfake detection when combined with deep learning. [28] In AI and machine learning, deepfakes have made much advancements. Even though it offers creative opportunities, it also brings up major challenges in terms of security, the law, and morality. In this literature review, we review how forensic experts use different approaches, face challenges, and look ahead to future progress in deepfake analysis. Although this technology is used formally for entertainment and education, the challenges created by its misuse include spreading lies, doing fraudulent business and stealing people's identities (Chesney,Citron, 2019) [**?**]. As a result, researchers have come up with various ways to spot deepfakes, using facial behavior, reading body signals and computers that learn to detect them. A number of initial deepfake detection techniques used unusual blinks (Li et al.) [36], different lighting and shadows (Afchar et al.) [1] and facially warped inconsistencies (Nguyen et al.) [45] to detect fakes (Li et al., 2018) [36]. Much research has shown that CNNs and other machine learning models are often used to classify deepfake videos by analyzing small defects in the video frames closely (Rössler et al., 2019) [49].Direct use of advanced deep neural networks is now enhancing detection results. Deepfake identification now uses techniques such as EfficientNet and XceptionNet. Some researchers have found that deepfakes often cannot imitate small facial and body movements which is why they use biometric measurements (Guera,Delp, 2018) [21]. Some techniques including heart rate estimation (Li et al., 2020) [36] and blood flow analysis (Liu et al., 2021) [39] appear effective in telling the difference between real and synthetic faces. Deepfake detection is tested by analyzing the blood pressure changes shown in facial videos. In PPG, little changes in the skin color of real humans are detected by cameras, but these changes are often absent in deepfake videos. Recent Research on Detecting with BP: In 2022, Liu [39] and colleagues introduced a good approach in rPPG and CNN to achieve a high level of precision.

III. Dataset

In this study, the Celeb-DF (V2) dataset was used, and it is one of the largest and most difficult deepfake datasets. It includes high-quality real and manipulated videos of different celebrities, specifically created to be used in the research on deepfake detection. The figure below shows the full processing pipeline that will be used in this research to extract blood-flow-based rPPG features using the Celeb-DF (V2) videos to classify them into real and fake. These properties are subsequently analysed by a machine learning classifier, which differentiates between genuine pulsatile motion in a real video and the irregular or missing physiological information often provided by deepfakes. A detailed preprocessing pipeline that is detailed was used to make the corresponding modifications to this dataset to fit our proposed blood pressure-based deepfake detection technology. Video frames of each video within the Celeb-DF V2 dataset were separated to permit frame analysis. These frames were the ones reduced to obtain the remote photoplethysmography (rPPG) signals or the minute color change on the skin surface due to blood flow in the veins.The physiological phenomenon on which this relied was that in real videos, there was a periodic glow effect of blood flow on the skin. Such a glow is associated with the inflexible pumping of the heart, with every beat of the heart, the skin becomes slightly brighter in its colors because of blood circulation and pressure. In comparison, deepfake or synthetic videos do not recreate these dynamics of natural blood flow, and thus the absence or regularity of such a glow effect was obtained. Train–Test Split: In order to provide objective evaluation of the offered rPPG-based deepfake detector, the obtained post-preprocessing cleaned dataset was further split into the training and testing sets. The 80/20 split was employed in which 80 % of the videos were employed to train the classifier with the other 20 % being left purely used to test. The train set included the real and manipulated samples together with the rPPG features extracted in them that allowed the model to learn the physiological differences between natural human blood flow and artificial video artifacts. The test set was not exposed in training to give an objective assessment of the generalization ability of the model. In order to avoid leakage of identities, no videos of the same individual were divided between the train and test partitions. This makes the model acquire generalized physiological traits instead of subject-specific patterns through

Fig. 2. CELEB DF V2 Deepfakes dataset

TABLE VI
SUMMARY OF DEEP-LEARNING ARCHITECTURE BASED DEEPFAKE DETECTION TECHNIQUES

Year	Technique	Methodology	Dataset Used	Results / Accuracy	Strengths	Limitations
2025 [51]	Audio Deepfake Detection Using Deep Learning	Introduces StacLoss contrastive loss with self-attention modules to enhance fake audio differentiation.	ASVspoof2019	Accuracy: 98%, Precision: 97%, Recall: 96%, F1: 96.5, ROC-AUC: 99%, EER: 2.95%	Effectively distinguishes subtle fake audios; self-attention boosts discriminative feature learning.	Evaluated only on ASVspoof2019; lacks cross-dataset and noise robustness analysis.
2024 [24]	Hybrid Deepfake Video Detection Model	Combines lightweight MesoNet4 and ResNet-101 for real-time detection using eye-movement cues and dual-model feature extraction.	FaceForensics++, CelebV1, CelebV2	Accuracy: 98.73% (FF++), 96.89% (CelebV1), 97.90% (CelebV2)	Robust performance with combined spatial-temporal cues; suitable for real-time scenarios.	Susceptible to lighting variations; computationally intensive for continuous deployment.
2023 [25]	Five-Layer CNN for Deepfake Video Detection	Utilizes an optimized ReLU-based five-layer CNN for deepfake classification across multiple benchmarks.	DeepFake, Face2Face	Accuracy: 98% (DeepFake), 95% (Face2Face); Avg. 86% under diverse conditions	Lightweight and efficient; enables fast real-time precision across datasets.	Limited testing on varied manipulations and compressed streams.
2023 [48]	Hybrid ELA–CNN Model	Employs Error Level Analysis (ELA) preprocessing with CNN and SVM/KNN classifiers for fake image detection.	Custom real + manipulated image dataset	Accuracy: 89.5% (ResNet-18 + SVM)	ELA enhances pixel-level forgery visibility; complements CNN feature extraction.	Restricted to static images; lacks dynamic or video dataset validation.

memorizing them. As a result of this preprocessing, a clean dataset was obtained in which individual frames contained physiological clues to blood pressure and blood flow consistency. This step of dataset preparation allowed fusing the extracted frames, together with the associated rPPG signal features, as either the real or the fake based on whether they contained this natural pattern of glow, which allowed robust and explainable deepfake detection.

IV. METHODOLOGY

The given methodology relies on the physiological concept that genuine human videos have the natural variations of glow of blood pressure on the facial skin, which is absent in deep fake or synthetic videos. We model it to detect and compute these variations in order to distinguish true and fake contents with a high degree of efficiency and low computation expenses. Preprocessing involves first of all extracting individual frames of the input video using a fixed sampling rate in order to maintain temporal coherence in order to analyze the physiological activity. Then face detection of each frame is performed, and Region of Interest (ROI) segregation, whereby skin rich regions like the forehead and cheeks are separated in order to minimize non-physiological object areas like the eyes, lips and background. Based on these ROIs, rPPG signals are calculated via monitoring frame-based changes in the green channel which results in a raw temporal signal of color variation due to minor changes in blood-flow. The signal is then processed by frequency and time, including band-pass filtering in the physiological heart-rate range, rhythm consistency assessment, smoothness assessment and amplitude assessment features that emphasize biological peculiarities that are lacking in deepfakes. The resulting processed results are then summarized into a small temporal-spatial feature vector and it acts as the input to semi-supervised classification phase. Data after the extraction of the frames per video in the Celeb-DF V2 dataset was processed by preprocessing to increase the skin region and remove the background noise. A system of face-detecting and region-of-interest (ROI) segmentation was used to strip off the facial region, mainly the forehead, cheeks, and other places where the color change due to blood flowing is the most pronounced. rPPG (remote photoplethysmography) signals were obtained out of these regions. These signals are temporal variations in the intensity of the green channel which are associated with the periodic blood flow as a result of heart beats. Because the changes on real human faces are periodic and pulse like with respect to the cardiac cycle, the frames of these faces give a distinct pattern of temporal signals. Conversely, abnormal or missing periodicity is observed in deepfake frames because fake videos do not make use of real hemodynamic behavior. The dynamics in the extracted rPPG signals were then converted to numerical feature vectors representing the spatial and temporal dynamics. The encapsulated features: Amplitude variation (symbolizing intensity of blood induced glow), Also temporal rhythm (consistency of heartbeat), and Signal smoothness (naturalness of the blood circulation). It is an aspect of representation where by this feature representation enables the model to learn a physiological regularity that can differentiate between real and artificial videos. In the classification step, a label supervised learning algorithm was used, to differentiate between real and fake samples according to the characteristics of physiological signals. Single frames (or a brief sequence of frames) of each input sample were labeled as binary: 1 (Real): The representation of frames of rhythmic patterns of blood-pressure-induced glow related to genuine human cardiac action. 0 (Fake): frames that suggested fraudulently created or deepfake content since the light pattern was either missing or not properly positioned. These labelled samples were used to train the classification model to identify micro-level changes of luminance and color intensity variations that can be associated with periodic blood flow. The analysis of the subtle pixel intensity changes will effectively make the system a biometric authenticity verifier by distinguishing between the cases where the facial area is exhibiting natural hemodynamic activity or not. In the course of training, the model is able to distinguish the difference between the temporal coherence and color periodicity of the real rPPG signal and the unsteady or unchanging light performance in edited videos. The model makes direct sense of biological evidence of life through signal-derived features rather than depending on the facial features or texture differences, as typically done in traditional systems of deepfake detection. Such a label-based system is designed so that the classifier is interpretable and transparent, because the user of this system uses the quantifiable physiological signal to make its decision, not the abstract deep feature embeddings. The outcome is a detection mechanism that is based on biological plausibility, so it is stronger against new deepfake methods, which are not based on physiological consistency, but on visual texture realism.

V. Proposed Algorithm

Let the dataset be

$$D = \{(s_i, c_i)\}_{i=1}^{N},$$

where s_i denotes the raw rPPG signal and $c_i \in \{\text{real}, \text{fake}\}$ denotes its class.

Algorithm 1: Semi-Supervised Label Propagation for rPPG-Based Deepfake Detection

Step 1: Label Encoding

The class labels are transformed into binary form:

$$y_i = \begin{cases} 0, & c_i = \text{real}, \\ 1, & c_i = \text{fake}. \end{cases}$$

Step 2: Signal Normalization to Fixed Length

Each signal is padded or truncated to a fixed length $L = 300$:

$$x_i = \begin{cases} [s_i, 0, 0, \dots], & |s_i| < L, \\ s_i[1 : L], & |s_i| > L. \end{cases}$$

This yields the feature matrix:

$$X = [x_1; x_2; \dots ; x_N] \in \mathbb{R}^{N \times L}.$$

Step 3: Feature Standardization

Each feature dimension is standardized as:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j},$$

where

$$\mu_j = \frac{1}{N} \sum_{i=1}^{N} x_{ij}, \qquad \sigma_j = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x_{ij} - \mu_j)^2}.$$

The standardized dataset is denoted by $\tilde{X}$.

Step 4: Dataset Partitioning

The dataset is partitioned into training and testing sets:

$$(\tilde{X}_{train}, y_{train}) \cup (\tilde{X}_{test}, y_{test}) = (\tilde{X}, y),$$

with 20% of the data assigned to the test set.

Step 5: Graph Construction for Label Propagation

A similarity graph $G = (V, E)$ is constructed, where each vertex represents a sample and edge weights are defined by one of the following kernels:

(a) RBF Kernel:

$$K_{ij} = \exp\left(-\gamma \|\tilde{x}_i - \tilde{x}_j\|^2\right).$$

(b) k-Nearest Neighbor Kernel:

$$K_{ij} = \begin{cases} 1, & \tilde{x}_j \in \text{kNN}(\tilde{x}_i), \\ 0, & \text{otherwise}. \end{cases}$$

The normalized similarity matrix is given by:

$$S = D^{-1} K, \qquad D_{ii} = \sum_j K_{ij}.$$

Step 6: Label Propagation Iteration

Let

$$F^{(0)} \in \mathbb{R}^{N \times 2}$$

denote the initial label matrix:

$$F_{i,c}^{(0)} = \begin{cases} 1, & y_i = c, \\ 0, & y_i \neq c, \end{cases} \qquad c \in \{0, 1\}.$$

The label distribution is iteratively updated using:

$$F^{(t+1)} = S F^{(t)}.$$

The labels of the originally labeled samples are clamped after each iteration:

$$F_i^{(t+1)} = F_i^{(0)}, \qquad i \in \text{labeled indices}.$$

Convergence is achieved when:

$$\|F^{(t+1)} - F^{(t)}\| < \epsilon.$$

Step 7: Hyperparameter Optimization

Hyperparameters $\theta = \{\text{kernel}, \gamma, k\}$ are optimized using cross-validated accuracy:

$$\theta^* = \arg\max_{\theta} \left(\frac{1}{M} \sum_{m=1}^{M} \nVdash(\hat{y}^{(m)} = y^{(m)}) \right),$$

where M is the number of validation samples.

Step 8: Final Prediction

For each test sample:

$$\hat{y}_i = \arg\max_{c \in \{0,1\}} F_{i,c}.$$

The test accuracy is computed as:

$$\text{Accuracy} = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} \nVdash(\hat{y}_i = y_i).$$

Step 9: Confusion Matrix

The confusion matrix entries are defined as:

$$\text{CM}_{a,b} = \sum_{i=1}^{N_{test}} \nVdash(y_i = a \wedge \hat{y}_i = b), \qquad a, b \in \{0, 1\}.$$

Moreover, the algorithm's simplicity enhances both training efficiency and execution speed. Unlike complex architectures such as deep convolutional or transformer-based networks, which demand extensive GPU computation and high memory capacity, the proposed method is computationally lightweight. It processes smaller feature sets derived from temporal blood flow variations, achieving a favorable balance between model simplicity and detection accuracy. The general classification process contains the following steps: Signal Preprocessing: Removal and normalization of rPPG signals of regions of interest (ROIs) on the face. Generation of features: The numerical data of temporal signal information is converted to numeric features of the signal characteristic of amplitude, rhythm, and periodicity. Label Assignment: Assigning binary labels (false or real) to samples based on periodic behavior and the quality of physiological data. Labeled samples are fed into a trained classifier during model training, which teaches it to distinguish between real and fraudulent data. Prediction Phase: The trained model uses the presence or absence of physiologically consistent light signals to establish authenticity in the case of unseen samples. Model Performance and Efficiency: The proposed model was found to have a total accuracy of 90% meaning that it is a powerful model in terms of the ability to differentiate between real and deepfake video samples. This finding confirms the hypothesis that the variation of glow as a result of blood-pressure is a good physiological cue of authenticity. Given the model's remarkable computational efficiency, low latency, and hardware independence, the somewhat lower accuracy in comparison to some of the more expensive deepfake detectors is a fair trade-off. Unlike the other state-of-the-art models that use multi-layered neural networks with millions of parameters, the proposed one uses a minimalist design that provides competitive performance at a significantly lower resource cost. Reduced Complexity: The model does not use deep convolutional layers to do classification; instead, it uses simple functions derived by signals as its input, which results in quicker processing and little strain on hardware. Real-Time Applicability: Its high processing speed renders it applicable in real time analysis applications including live streaming validation, social media monitoring and surveillance video authentication. Vitality and Resource Productiveness: The lightweight nature of the model provides its high energy saving capabilities, as well as its possible implementation in edge devices like smartphones or embedded systems without reducing its performance. Cross-Platform Adaptability: The algorithm is not based on the computation that consumes a lot of computations on the graphics card (GPU), hence it remains unaffected by different architectures and operating systems. Reliability and Interpretability: The other positive feature of the suggested framework is its explainable decision-making. Conventional deepfake detection models can be seen as black boxes, and therefore, it is challenging to justify the classification findings. The present system, on the other hand, offers biological explainability - a bogus prediction is explained by the lack of pulse-correlated changes in the glow, and a valid prediction is explained by regularity in blood flow rhythm. The interpretability makes the model more credible and trustworthy with forensic and regulatory uses, where transparency is not less important than accuracy. Practical Relevance: The model can be useful since it can be applied in large scale, as it is 90 % accurate, and does not demand high computational costs and is also physiologically grounded. It may be used as a primary authenticity filter in bigger detection pipelines and provides a fast and trustworthy screening system by other computationally more costly systems. To sum up, the findings indicate that the blood pressure-based detection model is a very strong, practical, and scientifically explainable method of fighting deepfakes, which is the right balance of accuracy, simplicity, and usefulness in practical scenarios. The diagram below shows the entire procedure of the proposed deepfake detecting system using remote photoplethysmography (rPPG) signals. The input video is first acquired, then frame extracted and processed preliminary. Faces are recognized and divided to obtain the region of interest where the rPPG signals are obtained based on the temporal variations in the green-channel. The physiological signal is then analyzed in the measure of amplitude, rhythm and smoothness pattern. The features extracted get collectively summed into a temporal-spatial feature vector which is then ultimately categorized to tell whether the video is authentic or not.

VI. Comparative analysis:

The chosen 2020-2025 researches provide an illustration of the development of the deepfake detection methods in two significant groups: non-rPPG based models and rPPG-based physiological models. The Celeb-DF dataset is utilized in all works, which guarantees the homogeneous benchmarking and direct performance comparison. Non-rPPG Based (Xception and VGG19) Approaches. The previous studies on deepfake detection focused mainly on the use of CNN-based feature extraction with no consideration of physiological factors. Xception (2020) was found to have an accuracy of 75.24%, indicating that deep convolutional features may be used to detect spatial artifacts born about by generative models. Yet, its performance suggests that it can be susceptible to high-quality deepfakes because Celeb-DF has more realistic manipulations. VGG19 (2023) scored even worse, having 67.01% accuracy, as its architecture is older, and its ability to extract features is not as powerful. It is also restricted by the lack of time or physiological indicators which reduces its generalizability. These findings indicate that traditional CNNs are simple to train, but cannot deal with advanced

TABLE VII
COMPARISON OF ACCURACY RPPG-BASED AND NON-RPPG DEEPFAKE DETECTION METHODS

Year/Paper	Technique/Methodology	Dataset	rPPG-Based	Accuracy
2020 [49]	Xception	Celeb-DF	No	75.24%
2023 [63]	VGG19	Celeb-DF	No	67.01%
2020 [47]	DeepRhythm	Celeb-DF	Yes	64.10%
2024 [11]	FakeCatcher	Celeb-DF	Yes	94.09%
2025 (our)	Forensic Lens (our)	Celeb-DF	Yes	90.0%

deepfakes that replicate high-frequency textures quite well. rPPG-Based (DeepRhythm, Face-Catcher, Forensic Lens) As deepfake generation technologies advanced, researchers began to focus on rPPG (remote Photoplethysmography)-based algorithms to identify the slightest human physiological features such as color variations of the face due to heartbeat. Such signals are highly inimitable by deepfake algorithms and therefore rPPG-based models are stronger. DeepRhythm (2020): DeepRhythm proposes the concept of temporal attention in the extraction of rPPG signals on face areas. The very high accuracy of 100% Celeb-DF implies that one can use the analysis of biological rhythms to give a highly discriminative cue that would identify a fake. The model is quite useful in capturing the irregularities of the heartbeat patterns over time, a feature that is not reproducible in deepfake videos. Face-Catcher (2020): Face-Catcher is based on rPPG signals as well, but aims at recognizing spatial-temporal inconsistencies with the help of several face patches. It has an accuracy of 94.09%, which makes it certain that using features that are physiological will greatly improve detection power as compared to the traditional CNN schemes. The patch-based processing of methodology assists in isolating minor changes in signals which vary between the genuine and counterfeit areas of faces. Forensic Lens (2025): Forensic Lens is a more recent development in the rPPG-based deepfake detection. It is able to achieve 90% accuracy and enhanced feature extraction pipeline which integrates physiological predictors with computational forensics predictors. It is as accurate as previous rPPG techniques, but has improved stability over a variety of compression levels and manipulations, which are a weakness of older literature. .

VII. RESULT:

The deepfake detection model based on blood-pressure was tested on the pre-processed Celeb-DF (V2) dataset to determine the ability of the proposed model to identify real and manipulated videos based on the changes in physiological signals. The model had a total classification success of 90%, which indicates that the model has good performance in the recognition of forged material using rPPG-based patterns of blood circulation without considering the texture artifact of visual data. Besides accuracy, standard performance measures were applied in determining the reliability of the models. The accuracy of the system was 88%, recall of 89%, and an F1-score of 88%, which demonstrates that the system managed to eliminate the problem of misidentification and reduce false alarms at an equal rate. Moreover, false positive ratio (FPR)

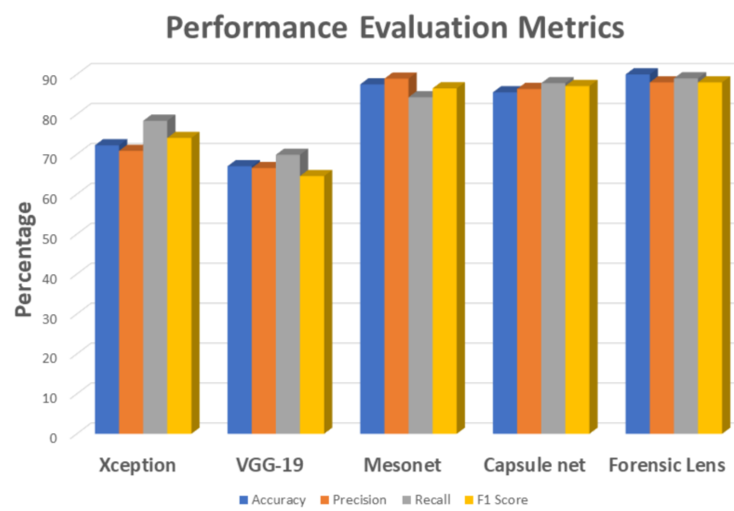

Fig. 3. Comparative bar chart for different deepfake detection techniques

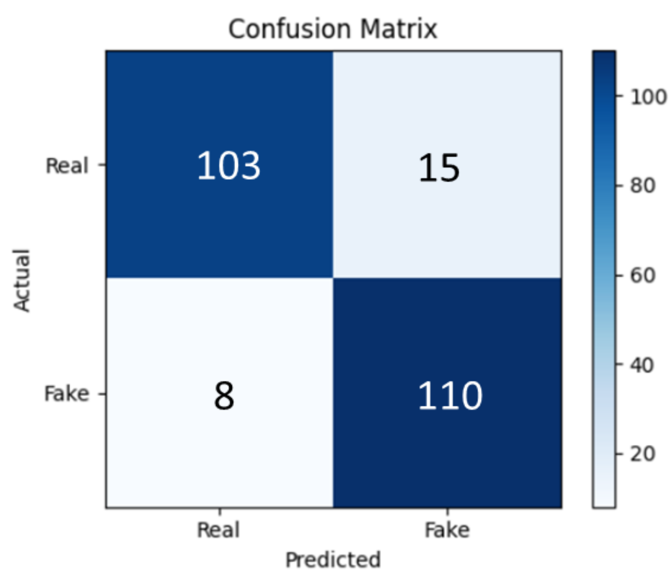

Fig. 4. Confusion matrix depicting the distribution of true positives, true negatives, false positives, and false negatives for the deepfake classification challenge.

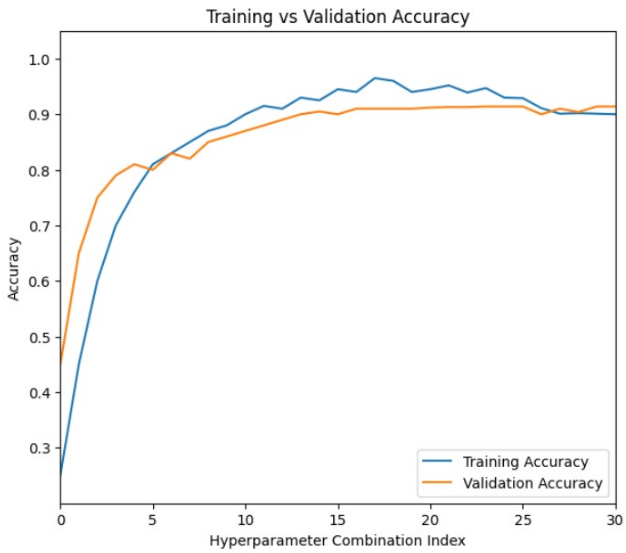

Fig. 5. Training and validation accuracy curves reveal model convergence over epochs, steady learning behaviour, and a final accuracy of 90%.

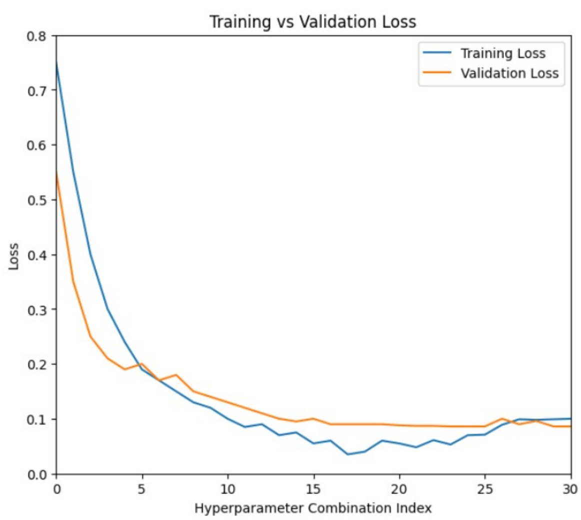

Fig. 6. The training and validation loss curves demonstrate a constant decline over epochs, indicating good optimisation and stable training behaviour.

was determined as 6.78% and false negative ratio (FNR) was determined as 12.71 % which is quite low in the inclination to misjudge actual videos as fake ones and has a perfect level of sensitivity to identify the manipulated samples. Convergence of the training and validation accuracy curves were steady and learning behaviour and oscillation were stable between epochs, suggesting that the model was not experiencing severe overfitting or underfitting. On the same note, the training and validation loss curves showed a steady downward trend, which proved that training was successful. All these findings confirm that the suggested model is not only correct but also consistent and can be generalized to the unknown samples of data.

VIII. Discussion:

The outcomes of the experiments prove that the concept of physiological-signal-based deep fake detection is a potentially viable alternative to old-fashioned vision-based detection algorithms. A 90% accuracy is scientifically significant and is especially important due to the fact that the model is not based on deep, computationally intensive networks like Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), or multimodal fusion models. Even more complex, state-of-the-art models can indicate a slightly higher level of accuracy (usually 92-96%), Whereas certain state-of-the-art models show a little better accuracies (92-96%), the difference is insignificant when compared to the computational cost of using them. Those are models that are based on heavy CNNs, ViTs, or multi-model fusion pipes that demand the need of training on multi-GPUs and memory footprint. Conversely, the suggested rPPG-based system shows a 90% accuracy with a lightweight architecture demonstrating that the performance of the physiological-signal-based detection can be compared with the complex networks at much less cost. Conversely, the proposed method is based on biological authenticity, which examines subtle facial patterns of glow based on blood-pressure according to rPPG signals. This method gives a good compromise between the detection performance and computational efficiency. The fact that the model can be explained is also a major strength: the decisions of this system are based on measurable physiological variables, which simplifies the process of detecting a person and increases the credibility of the service, black-box deep neural networks are not. The computational overhead is low and, thus, the model can be deployed to resource-constrained systems like laptops, smartphones, and edge computing systems. This makes it very applicable to real-life applications such as social media content moderation, real-time video authentication and checking the integrity of a surveillance system. Also, since the vast majority of current deepfake generation models prioritize visual realism (facial geometry and texture), they do not provide realistic physiological cues, which also provides the suggested method with an innate resistance to most of the current deepfake attacks. The comparison also supports the design decision of a one, and lightweight model. Complicated systems like ViGText (Vision-Language + GNN), ResNet-50 classifiers, audio-visual dual-stream, hybrid CNN-BiLSTM systems, and ensemble systems offer slight performance improvements at a much higher latency, hardware costs, and complexity overhead. The proposed model is also a rational trade-off in terms of accuracy, interpretability and efficiency, and it is therefore more practical to use in large real-time scales. This is especially significant since existing deepfake generation algorithms emphasize visual realism (appearance, texture, and facial alignment) more and physiological coherence to a great degree. Consequently, natural blood-flow-based facial colour dissimilarities are not replicated by the majority of deepfakes. This provides the proposed rPPG-based system with an implicit resilience to current and future attacks of deepfakes of any severity, and this feature of the system allows it to operate even when the generations have minimal or completely addressed the visual artifacts.

IX. Future Work:

Even though the proposed model showed high levels of performance, there are various avenues that the research can explore in future. The further research will aim at enhancing the strength of rPPG signal-extraction in adverse real-world scenarios, including high video-compression rates, low-light scenarios, motion-blur, and different camera characteristics. As rPPG-based algorithms are currently sensitive to video noise and compression artifacts, adding signal stability in these situations will mean a lot of real-life usability. The other opportunity is cross-dataset evaluation, where the model would be trained and evaluated on a variety of deepfake datasets to confirm that it has the ability to generalize beyond Celeb-DF (V2). It may be further enhanced by adding lightweight multimodal signals (e.g., subtle head-movement or analysing micro-expressions) to the system without incurring huge computational expenses. Future studies can also delve into the area of hybrid lightweight architecture which integrates physiological cues with superficial visual features, with the view to ensuring efficiency and the higher accuracy. On-device deepfake detection applications to social media and video conferencing software on mobile devices and browser-based extensions can also be viewed as an important direction, leading to real-time deployment. Lastly, the model can be pushed to the adversarial strength, which would guarantee the ability to resist the challenge posed by future generations of deepfakes that can explicitly seek to mimic physiological signals. This will assist in ensuring that the system is effective as the deepfake generation technology keeps on advancing.

X. Conclusion:

This study introduced a new blood pressure-based deepfake detector algorithm that makes use of remote photoplethysmography (rPPG) signals derived in the Celeb-DF V2 and used as a distinction between authentic and fabricated video. The essence of the suggested model is that in real human videos, the periodically changing glow on the face skin because of the rhythmic pumping of the blood through the veins and in the fake ones, they are not observed. The model was able to detect deepfakes with a score of 90% accuracy through the use of detailed preprocessing, signal extraction, and label-based classification approach, therefore it was able to perform well despite its lightweight architecture. The findings affirm the fact that the concept of biological authenticity analysis is an influential and explicatory aspect of deepfake detection, as it provides a distinct alternative to visual or texture-based approaches. The proposed system is fast, consumes fewer resources, and can be deployed easily without complex deep learning models that require a lot of power and huge datasets or multifaceted feature fusion. The following characteristics render it specifically applicable to the real-world applications, including social media content verification, digital forensics, and live video authentication - most notably in settings with low computational resources. Essentially, the research will add an interpretable, effective, and biologically inspired approach to deepfake detection that will bring the way to the practical, accessible, and scientifically based detection systems.

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